



Practical - 29

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Class : SYIT	Date/Time of Submission : 22-09-2026

```
// 29) Create a simple Spark MLlib pipeline to  
classify data. Use logistic regression or decision  
trees.
```

Code:

```
// Build.sbt  
// libraryDependencies += Seq(  
//   "org.apache.spark" %% "spark-sql" % "3.5.9",  
//   "org.apache.spark" %% "spark-mllib" % "3.5.9"  
// )  
  
// Demo.scala  
import org.apache.spark.sql.SparkSession  
import org.apache.spark.ml.feature.VectorAssembler  
import org.apache.spark.ml.classification.LogisticRegression  
  
object Prac10 extends App {  
  
  val spark = SparkSession.builder()  
    .appName("Classification")  
    .master("local[*]")  
    .getOrCreate()  
  
  import spark.implicits._  
  
  val data = Seq(  
    (1.0, 0.0),  
    (2.0, 0.0),
```

```

    (3.0, 0.0),
    (7.0, 1.0),
    (8.0, 1.0),
    (9.0, 1.0)
  ).toDF("marks", "label")

val a = new VectorAssembler()
  .setInputCols(Array("marks"))
  .setOutputCol("features")

val d = a.transform(data)

val model = new LogisticRegression()
  .setLabelCol("label")
  .setFeaturesCol("features")

val result = model.fit(d)

val prediction = result.transform(d)

prediction.select("marks", "label", "prediction").show()

spark.stop()

println("Amit 44")
}

```

Output:

```

+-----+-----+-----+
|marks|label|prediction|
+-----+-----+-----+
|  1.0|  0.0|         0.0|
|  2.0|  0.0|         0.0|
|  3.0|  0.0|         0.0|
|  7.0|  1.0|         1.0|
|  8.0|  1.0|         1.0|
|  9.0|  1.0|         1.0|
+-----+-----+-----+

```

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Process finished with exit code 0